

A Group-Stratified Reliability Framework for Trustworthy Early-GPA Prediction in Higher Education

Sayuri Witharana
Faculty of Business Science
University of Moratuwa
Moratuwa, Sri Lanka
witharanassb.22@uom.lk

Tiloka de Silva
Faculty of Business Science
University of Moratuwa
Moratuwa, Sri Lanka
tilokad@uom.lk

Abstract— Student performance shows significant variability in the higher education context. Early forecasting of Grade Point Average (GPA) will help identify students who are performing well and those who are struggling academically, which would facilitate the provision of additional assistance for these students. Existing frameworks for determining the earliest prediction semester mainly focus on model accuracy which ignores the prediction stability and systematic bias toward students at the lower tails of the distribution. Using data on undergraduate students from a Sri Lankan State University from 2017 to 2019, this research provides a novel three-criterion reliability framework that addresses prediction accuracy, prediction stability, and group-level directional bias. Results indicate that the point at which predictions become reliably actionable varies meaningfully across student performance groups, driven primarily by convergence of directional bias and reductions in Root Mean Squared Error (RMSE) variability. The proposed methodology is transferable across institutional settings and explicitly accounts for student heterogeneity, providing a structured basis for responsible, equity-aware academic intervention.

Keywords— GPA prediction, education analytics, prediction reliability, algorithmic bias in education, prediction stability, higher education

I. INTRODUCTION

Predictive analytics in higher education has driven widespread use of machine learning for predicting student academic outcomes [1]. Early GPA prediction helps institutions promptly identify students at-risk and students who are high achieving [2]. In digital learning environments, these predictions are built into automated advising systems, where inaccurate estimates could become institutional facts applied at scale without direct human oversight.

Model evaluation in these contexts remains largely accuracy-based, overlooking a significant inequity: a model accurate for most students can be over-optimistic for low performers, leading to false security [3]. In practice, consistency across samples and directional bias is rarely evaluated, making widespread deployment questionable [4]. This study introduces a reliability framework centred on the stability of GPA prediction, alongside accuracy, and directional bias, to identify the earliest semester at which predictions become reliably actionable for each student group.

II. RESEARCH QUESTION

The study addresses the following primary research question:

What is the earliest semester at which a student's final GPA can be forecasted reliably across different student performance groups, accounting for prediction stability and error asymmetry across student groups?

To answer this, the study identifies the earliest semester satisfying accuracy, stability, and equity thresholds per group, and determines which reliability criterion most constrains the onset of trustworthy prediction.

III. LITERATURE REVIEW

GPA prediction has been framed mainly as classification. Of the 24 studies reviewed, 75% used categorical outcomes such as pass/fail or degree class [5], which makes error magnitude hard to assess and impossible to quantify continuous, group-level bias.

A deeper gap prevails in the reliance on single-point accuracy metrics. Across 212 explainability-focused studies, only 3.3% reported stability indicators and 5.6% reported fairness metrics [6]. Classification work often reports near-perfect recall for high performers but poor recall for low performers [7], yet this asymmetry is rarely quantified in regression settings [8].

Early-warning research stresses that a delayed or inaccurate alert reduces intervention effectiveness [2] and that timing matters as much as accuracy [9]. There is also a need for group-level evaluation from fairness-in-machine-learning frameworks [3,4]. Collectively, the studies reveal lack of a single reliability criterion in the GPA prediction literature integrating stability and equity with accuracy.

IV. METHODOLOGY

A. Dataset and Temporal Design

Data were obtained from a business faculty at a Sri Lankan state university over three academic years (2017-2019): 271 records, with 178 for training and the 2019 data as an external test set. Features include pre-academic factors (Z-score, English marks and medium of Advanced-level examinations), demographics (gender, department, district), and academic features (semester GPA). Students were categorized by final GPA as High-performing ($3.3 \geq$), Average ($3.29 - 3.0$), or Underperforming (≤ 2.99). These labels were used for stratification only.

B. Model Selection

Linear regression and Random Forest were initially selected due to their predominance in GPA prediction literature. Linear regression was implemented with L2 regularization (ridge) to address multicollinearity between semester GPAs and to preserve the stability of prediction outputs given small sample [10]. Ridge regression showed output stability and lower directional bias compared to Random Forest and was therefore chosen for the analysis.

C. Evaluation Framework

Prediction reliability was evaluated using a two-stage process. Stage-1 applies a three-criteria framework across repeated stratified cross-validation (5-fold, 10 repetitions, 50 splits total), using the distribution of RMSE across the 50 splits to assess prediction sensitivity to data composition - not for model tuning. Proportional subgroup representation was preserved through stratification across all splits. Stage-2 evaluates a ridge regression model trained on the full 2017–2018 dataset against the 2019 dataset to confirm those candidates.

A semester is classified reliably actionable when all criteria are satisfied:

- (1) Accuracy: $RMSE \leq 0.25$ for High-performing and Average groups; ≤ 0.35 for Underperforming.
- (2) Stability: Evaluated by comparing Standard Deviation (SD) of RMSE across semesters; Must demonstrate a convergent trend.
- (3) Equity: $|\text{Group bias}| \leq 0.10$ GPA points, constraining systematic directional error.

Thresholds were grounded in the 4-point GPA scale; 0.25 represents half of a practically meaningful grade band (0.5 GPA points) and the bias of 0.10 represents the point at which directional error consumes 40% of acceptable RMSE.

V. FINDINGS

Fig. 1 shows the convergence of the three reliability criteria across semesters for each performance group.

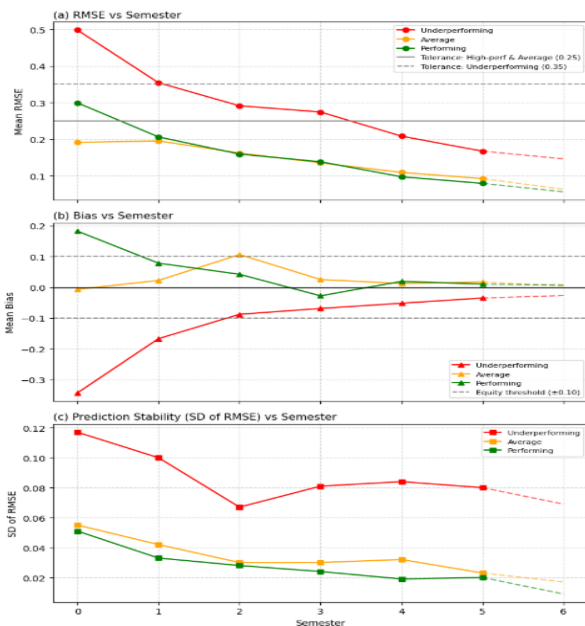


Fig. 1 CONVERGENCE OF THE THREE RELIABILITY CRITERIA

Table II shows the external test-set results of Stage-2 of the analysis.

TABLE I. TEST RESULTS (2019)

Semester	High performing		Average		Underperforming	
	RMSE	Bias	RMSE	Bias	RMSE	Bias
Sem 2	0.208	-0.152	0.378	-0.321	0.603	-0.480
Sem 3	0.142	-0.087	0.228	-0.149	0.426	-0.302
Sem 4	0.091	-0.017	0.144	-0.072	0.305	-0.202
Sem 5	0.072	0.048	0.094	0.041	0.163	-0.035

For High-performing and Average groups, all criteria passed at Semester 2 and Semester 3 respectively in Stage-1 cross-validation, as shown in Fig. 1. However, external validation revealed bias exceedance at both candidate semesters, advancing the confirmed reliability point by one semester in each case. Semester 3 and Semester 4 are thus identified as the externally validated reliability points, demonstrating that bias-based calibration failures remain undetected under accuracy-only evaluation.

Based on Stage-1 cross-validation, RMSE and bias for Underperforming students were satisfied from Semester 3; however, SD(RMSE) declined only briefly before rising through Semesters 3 and 4, and fell again only at Semester 5, remaining comparatively higher throughout than the other two groups. External test results confirmed RMSE and bias were satisfied only at Semester 5, establishing it as the conditional earliest reliability point for this group; conditional due to the persistently elevated SD. SD(RMSE) was the binding constraint throughout, reflecting the inherent prediction difficulty of this group (n=46). The graph shows Underperforming students were substantially overpredicted in early semesters, with the overprediction narrowing sharply and becoming negligible by Semester 5.

VI. DISCUSSION

Two key findings hold significant insights for academic support systems. First, reliable prediction semesters differ among the student performance groups. Second, accuracy alone is an insufficient deployment criterion; RMSE met the thresholds prior to the bias and stability criteria. This is particularly consequential for Underperforming students, whose predictions do not become fully reliable until Semester 5. Rather than withholding support until then, pre-Semester 5 predictions can be adjusted using the semester-specific bias estimates reported in Fig. 1 and used by advisors as a directional, supplementary signal for early support planning; By Semester 5, both bias and stability fall within tolerance, allowing predictions to be used with full confidence.

This study is scoped to demonstrate that reliable GPA prediction differs meaningfully across student performance groups, not to identify the optimal predictive model; hyperparameter tuning and broader model exploration are accordingly outside its bounds, since the goal is comparability of a fixed-specification model across semesters and groups rather than maximizing accuracy. Findings are based on a single-institution dataset.

As Stage-1 and Stage-2 reliability for the Underperforming group is only confirmed by Semester 5, a trustworthy numeric prediction alone cannot support timely advising for this group. The future phase of this work therefore shifts from prediction to explanation: incorporating module-level grades to capture semester-specific academic signals beyond cumulative GPA, testing whether individual features act as statistically consistent negative early-warning signals for eventual underperformance as early as Semester 0–2, and tracking how feature importance evolves across semesters to explain why reliability improves when it does. Together, these allow flagging academically meaningful risk factors well before Semester 5–6, even while the numeric prediction itself remains provisional for that group, supporting early academic advising.

VII. CONCLUSION

This study offers a three-criterion reliability framework to gauge when GPA predictions become reliably actionable for differing performance groups. Results indicate that prediction reliability is group-dependent, with directional bias and stability identified as the main constraining factors. A key limitation of this study is its small overall training sample ($n=177$), which is further reduced once split by performance group: the Underperforming subgroup contains only 46 students, meaning each cross-validation fold trains on roughly nine of them. This small fold size makes stability estimates highly sensitive to exactly which students land in which fold, and is the likely source of the non-monotonic SD(RMSE) trend observed for this group; the Underperforming reliability timeline reported here should be treated as a preliminary estimate, not a precise threshold, until validated on a larger cohort. This constraint applies to the specific findings from this cohort, not to the method itself: the three-criterion evaluation is a reusable, equity-focused framework that any institution can reapply to its own, ideally larger, dataset to obtain more stable, group-specific reliability estimates.

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